

BORAMA HARDWARE

Tech Stack & Dependency Specification

Every Library • Every Version • Every Config File

Next.js 14 • Supabase • Vercel • Google Gemini API

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Field	Value
Document type	Tech Stack & Dependency Specification
Project	Borama Hardware Store
Package manager	pnpm 9.12.1
Node.js runtime	20.18.1 LTS (Jod)
Framework	Next.js 14.2.15 (App Router)
Database	Supabase (PostgreSQL 15 + pgvector + RLS)
Hosting	Vercel (Pro recommended)
AI	Google Gemini 1.5 Pro / Flash + text-embedding-004
Total packages	38 production + 15 development = 53 pinned versions
Claude Code	Optimised — all packages have 10k+ training examples

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1. Selection Criteria & Philosophy

Every package in this document was chosen against a strict set of rules. The goal is a stack with maximum Claude Code compatibility — meaning Claude has seen thousands of real-world examples of every library during training, so generated code is accurate on the first attempt with minimal hallucination.

Rule	Requirement	Exceptions Allowed?
Age	Library must have been publicly available for 2+ years	No
Popularity	Library must have 10,000+ GitHub stars	Yes — if no established alternative exists (documented per entry)
Maintenance	Repository must have a commit in the last 12 months	No
TypeScript	Library must ship TypeScript types (built-in or @types)	No
Compatibility	All packages must support Node.js 20 LTS	No
Versioning	Every dependency pinned to an exact version (no ^ or ~)	No

Claude Code compatibility note: Next.js 14 App Router (not Next.js 15) is chosen deliberately. The majority of Claude's training data covers Next.js 13/14 App Router patterns. Next.js 15 introduced async params/cookies and changed several APIs — Claude generates incorrect code for these changes more often. Next.js 14.2.x is stable, production-proven, and produces the most accurate Claude Code output.

2. Runtime & Build Tooling

2.1 Node.js — v20.18.1

Node.js 20 is the Active LTS release (maintenance until April 2026). All packages in this stack are validated against Node 20. Pin the version using `.nvmrc` and enforce it in `package.json` engines.

Property	Value
Version	20.18.1
Release line	Node.js 20 LTS (codename 'Jod')
End of life	April 30, 2026
V8 engine	11.3
npm bundled	10.8.2 (not used — pnpm is the package manager)
GitHub stars	107k 

Enforce in `package.json`:

```
"engines": {
  "node": ">=20.18.1 <21.0.0",
  "pnpm": ">=9.12.1"
}
```

2.2 pnpm — v9.12.1

pnpm is the package manager. It is faster than npm, uses content-addressable storage to avoid disk duplication, and produces strict, reproducible installs. Claude Code has strong pnpm knowledge — all install commands in this document use pnpm.

Property	Value
Version	9.12.1
Install	corepack enable && corepack prepare pnpm@9.12.1 --activate
Lock file	pnpm-lock.yaml (commit to git, never hand-edit)
GitHub stars	29k 

3. Application Framework

3.1 Next.js — 14.2.15

Property	Value
Package	next
Version	14.2.15
Router	App Router (app/ directory). Pages Router not used.
Rendering	SSR for public pages, RSC for data-heavy admin pages, client components for interactive UI
API routes	app/api/** Route Handlers (not pages/api/)
Image opt.	next/image with Sharp (server-side)
Font opt.	next/font with Inter (Google Fonts, self-hosted at build time)
Released	October 2023
GitHub stars	128k 

3.2 React — 18.3.1 & React DOM — 18.3.1

Package	Version	Purpose
react	18.3.1	Core React library. Concurrent features, Suspense, useTransition.
react-dom	18.3.1	DOM renderer. Required by Next.js for client component hydration.

Install command:

pnpm add next@14.2.15 react@18.3.1 react-dom@18.3.1

4. Language — TypeScript

Property	Value
Package	typescript
Version	5.6.3
Mode	Strict mode enabled (strict: true in tsconfig.json)
Path aliases	@/* maps to ./src/*
Module res.	bundler (required for Next.js 14 App Router)
JSX	preserve (Next.js transforms JSX at build time)

Property	Value
Released	2012 (TypeScript 5.x series: 2023)
GitHub stars	100k 

Install command:

`pnpm add -D typescript@5.6.3 @types/node@20.17.6 @types/react@18.3.12 @types/react-dom@18.3.1`

5. Database & Authentication — Supabase

Supabase is the backend-as-a-service layer. It provides PostgreSQL 15, Row Level Security (RLS), Auth (phone OTP), Realtime websockets, Object Storage, and the pgvector extension for AI embeddings — all in one managed platform.

Component	Package / Service	Version / Detail
JavaScript client	@supabase/supabase-js	2.45.4
SSR helpers	@supabase/ssr	0.5.1 — server-side session management in Next.js App Router
PostgreSQL	Managed by Supabase	PostgreSQL 15 (no version choice — Supabase managed)
pgvector	Supabase extension	0.7.0 — enable via Supabase dashboard SQL editor
pg_cron	Supabase extension	Managed — enable via Supabase dashboard for scheduled jobs
Auth	Supabase Auth	Built-in. Phone OTP via Twilio. No separate package.
Realtime	Supabase Realtime	Built-in. Part of @supabase/supabase-js client.
Storage	Supabase Storage	Built-in. Product images and receipts.
GitHub stars	@supabase/supabase-js	13k 

5.1 Supabase Client Configuration

Two clients are required for Next.js App Router:

- **Browser client** (src/lib/supabase/client.ts) — used in Client Components ('use client')

```
import { createBrowserClient } from '@supabase/ssr'
export function createClient() {
  return createBrowserClient(
    process.env.NEXT_PUBLIC_SUPABASE_URL!,
    process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!
  )
}
```

- **Server client** (src/lib/supabase/server.ts) — used in Server Components, Route Handlers, Server Actions

```
import { createServerClient } from '@supabase/ssr'
import { cookies } from 'next/headers'
export function createClient() {
  const cookieStore = cookies()
  return createServerClient(
    process.env.NEXT_PUBLIC_SUPABASE_URL!,
    process.env.NEXT_PUBLIC_SUPABASE_ANON_KEY!,
    { cookies: { getAll() { return cookieStore.getAll() },
    setAll(c) { c.forEach(({ name,value,options }) =>
    cookieStore.set(name,value,options)) } } }
  )
}
```

```
)  
}
```

Install command:

```
pnpm add @supabase/supabase-js@2.45.4 @supabase/ssr@0.5.1
```

Required Supabase extensions (enable in SQL Editor):

```
create extension if not exists vector; -- pgvector for AI embeddings
```

```
create extension if not exists pg_cron; -- scheduled forecast jobs
```

```
create extension if not exists unaccent; -- for Somali text search normalisation
```


6. Hosting & Deployment — Vercel

Vercel is the hosting platform. No versioned package — deployment is configured via `vercel.json` and environment variables set in the Vercel dashboard.

Property	Value
Plan	Pro (required for: custom domains, team access, longer function timeouts, more bandwidth)
Build command	<code>pnpm build</code>
Install command	<code>pnpm install --frozen-lockfile</code>
Output directory	<code>.next</code> (auto-detected)
Node.js version	20.x (set in Vercel project settings → General → Node.js Version)
Functions region	<code>fra1</code> (Frankfurt) — closest to Somaliland via Europe
Edge functions	Used for: auth middleware, rate limiting (Upstash Redis)
Serverless timeout	Pro plan: 60s max. AI routes (Gemini) need up to 30s.
Environment vars	Set in Vercel dashboard. Never commit <code>.env</code> to git.

`vercel.json` (root of project):

```
{
  "buildCommand": "pnpm build",
  "installCommand": "pnpm install --frozen-lockfile",
  "framework": "nextjs",
  "regions": ["fra1"],
  "functions": {
    "src/app/api/ai/**": { "maxDuration": 30 },
    "src/app/api/auth/**": { "maxDuration": 10 }
  }
}
```

7. AI & Machine Learning — Google Gemini API

Property	Value
Package	<code>@google/generative-ai</code>
Version	0.21.0
Provider	Google AI (not Vertex AI — simpler setup for this use case)
Models used	gemini-1.5-pro (complex reasoning: project estimator, inventory forecast)
	gemini-1.5-flash (fast, cheap: chatbot, AI descriptions, search)
	text-embedding-004 (vector embeddings for semantic search)

Property	Value
API key	GEMINI_API_KEY — server-side only, never exposed to browser
Embedding dims	768 dimensions (matches pgvector index in products table)
Rate limits	gemini-1.5-pro: 2 RPM / 32k TPM (free). gemini-1.5-flash: 15 RPM / 1M TPM (free)
Released	December 2023
GitHub stars	Official Google SDK 

Usage pattern for all Gemini calls:

```
import { GoogleGenerativeAI } from '@google/generative-ai'
const genAI = new GoogleGenerativeAI(process.env.GEMINI_API_KEY!)
// Fast model – chatbot, descriptions
const flash = genAI.getGenerativeModel({ model: 'gemini-1.5-flash' })
// Smart model – project estimator, forecasting
const pro = genAI.getGenerativeModel({ model: 'gemini-1.5-pro' })
// Embeddings – semantic search
const embed = genAI.getGenerativeModel({ model: 'text-embedding-004' })
```

Install command:

```
pnpm add @google/generative-ai@0.21.0
```

8. UI Layer — Tailwind CSS + shadcn/ui + Radix UI

8.1 Tailwind CSS — 3.4.14

Package	Version	Purpose
tailwindcss	3.4.14	Core utility-first CSS framework
autoprefixer	10.4.20	PostCSS plugin — adds vendor prefixes
postcss	8.4.47	CSS transform pipeline (required by Tailwind)
@tailwindcss/forms	0.5.9	Resets default browser form styles for Tailwind compatibility
@tailwindcss/typography	0.5.15	Prose styles for AI-generated product descriptions

Install:

```
pnpm add -D tailwindcss@3.4.14 autoprefixer@10.4.20 postcss@8.4.47 @tailwindcss/forms@0.5.9 @tailwindcss/typography@0.5.15
```

8.2 shadcn/ui — CLI (no versioned package)

shadcn/ui is not an npm dependency — it is a CLI tool that **copies component source code directly into your project** under `src/components/ui/`. You own every file. This means: no version lock issues, full customisation, and Claude Code can read and edit these files directly.

Property	Value
Initialise	pnpm dlx shadcn@latest init
Add component	pnpm dlx shadcn@latest add button (repeat per component)
Style	Default (Tailwind-based, no CSS variables theme required)
GitHub stars	73k  (over 2 years old — created Jan 2023)

8.3 shadcn/ui Dependencies (auto-installed by CLI)

Package	Version	Stars	Role
@radix-ui/react-dialog	1.1.2		Modal dialogs, drawers
@radix-ui/react-select	2.1.2		Dropdown selects
@radix-ui/react-tabs	1.1.1		Tab navigation in admin
@radix-ui/react-dropdown-menu	2.1.2		Context menus, user menus
@radix-ui/react-checkbox	1.1.2		Checkboxes (filters, bulk select)
@radix-ui/react-switch	1.1.1		Toggle switches

Package	Version	Stars	Role
@radix-ui/react-slider	1.2.1	■	Price range filter
@radix-ui/react-toast	1.2.2	■	Toast notifications
@radix-ui/react-avatar	1.1.1	■	Customer avatar in nav
@radix-ui/react-progress	1.1.0	■	Loyalty tier progress bar
@radix-ui/react-separator	1.1.0	■	Visual dividers
@radix-ui/react-label	2.1.0	■	Form field labels
@radix-ui/react-slot	1.1.0	■	Polymorphic component wrapper
@radix-ui/react-tooltip	1.1.3	■	Hover tooltips
@radix-ui/react-accordion	1.2.1	■	Collapsible product details
@radix-ui/react-scroll-area	1.2.0	■	Custom scrollbars in POS cart
class-variance-authority	0.7.0	■■ 5k	shadcn/ui button variant system
clsx	2.1.1	■■ 8k	Conditional className utility
tailwind-merge	2.5.4	■ 10k	Merge Tailwind classes safely
lucide-react	0.454.0	■ 11k	Icon library (all UI icons)

■■ **Star count exceptions:** class-variance-authority (5k) and clsx (8k) are core shadcn/ui dependencies with no substitutes in this ecosystem. Both are actively maintained, widely used by millions of projects, and are safe choices despite being under 10k stars.

9. Charts — Recharts

Package	Version	Stars	Purpose
recharts	2.13.3	■ 24k	Revenue line chart, top products bar chart, orders donut chart, inventory bar chart

Recharts is built on React + D3. Its declarative JSX API maps perfectly to React component patterns. Claude Code has extensive Recharts knowledge. Import only what you use — tree-shaking keeps bundle small.

```
import { LineChart, Line, XAxis, YAxis, CartesianGrid,
  Tooltip, ResponsiveContainer, BarChart, Bar, PieChart, Pie } from 'recharts'
```

Install: `pnpm add recharts@2.13.3`

10. State Management

10.1 Zustand — 4.5.5 (client-side global state)

Property	Value
Package	zustand
Version	4.5.5
Used for	Cart state, POS session state, language preference, active location
Pattern	One store per domain slice. No Provider wrapper required.
Devtools	zustand/middleware — devtools middleware for Redux DevTools
Persistence	zustand/middleware — persist middleware with IndexedDB storage adapter
Stars	■ 47k

10.2 TanStack Query — 5.59.0 (server state)

Package	Version	Purpose
@tanstack/react-query	5.59.0	Server state: products, orders, inventory — caching, background refresh, optimistic updates
@tanstack/react-query-devtools	5.59.0	Dev-only devtools panel (excluded from production build)

Install:

```
pnpm add zustand@4.5.5 @tanstack/react-query@5.59.0
```

```
pnpm add -D @tanstack/react-query-devtools@5.59.0
```

11. Forms & Validation

Package	Version	Stars	Purpose
react-hook-form	7.53.1	■ 41k	Performant form state. Minimal re-renders. Used on: checkout, product form
zod	3.23.8	■ 34k	TypeScript-first schema validation. Validates: form inputs, API request bo
@hookform/resolvers	3.9.1	■■ 5k	Connects React Hook Form to Zod. Required for z.object() schemas to w

Install:

pnpm add react-hook-form@7.53.1 zod@3.23.8 @hookform/resolvers@3.9.1

12. Internationalisation — next-intl

Property	Value
Package	next-intl
Version	3.22.1
Languages	English (en) and Somali (so). JSON message files in messages/en.json and messages/so.json.
Pattern	App Router i18n routing with [locale] segment. Middleware rewrites routes.
Server use	getTranslations() in Server Components and Route Handlers
Client use	useTranslations() hook in Client Components
Stars	■■ 8k — No well-established App Router alternative. react-i18next (9k) does not integrate natively with Next

Install: pnpm add next-intl@3.22.1

13. Offline & PWA — next-pwa

Property	Value
Package	next-pwa
Version	5.6.0
Stars	■■ 7.8k — No well-established alternative for Next.js PWA. Workbox (raw) requires manual config for Next.js
Powered by	Workbox 6 (Google's service worker library)
Caches	Customer store pages, product catalog API responses, static assets
Offline mode	Serves cached pages when network unavailable. Cart preserved in IndexedDB.
Config	next.config.js (see Section 24). Disable in development to avoid cache issues.

■■ **App Router compatibility note:** next-pwa 5.6.0 uses CommonJS configuration. Use next.config.js (not .ts or .mjs) to avoid ESM import errors. The service worker registers correctly in production. Set disable: true in development to prevent stale cache during development.

Install: npm add next-pwa@5.6.0

14. Rate Limiting — Upstash Redis

Package	Version	Stars	Purpose
@upstash/ratelimit	2.0.4	■■ 4k	Sliding window rate limiting for API routes. Serverless-compatible — no
@upstash/redis	1.34.3	■■ 4k	HTTP-based Redis client. Used by @upstash/ratelimit as the backing st

■■ **Star count exception:** Upstash is the industry standard for serverless rate limiting on Vercel Edge. Standard Redis clients (ioredis, node-redis) require persistent TCP connections which are incompatible with Vercel serverless functions. Upstash's HTTP client is the only production-grade solution for this architecture.

Rate limits applied in this project:

Route / Action	Limit	Window
POST /api/auth/send-otp	3 requests	Per phone number per hour
GET /api/search	30 requests	Per IP per minute
POST /api/ai/chat	20 requests	Per user per hour
POST /api/ai/estimate	10 requests	Per user per hour
POST /api/orders	5 requests	Per user per minute
POST /api/auth/verify-otp	10 requests	Per phone number per hour

Install:

```
pnpm add @upstash/ratelimit@2.0.4 @upstash/redis@1.34.3
```

15. Notifications — Twilio SMS

Property	Value
Package	twilio
Version	5.3.5
Stars	■■ 3k — Official Twilio Node.js SDK. Twilio is the industry-standard SMS provider.
Used for	Order confirmed, payment confirmed, ready for pickup, order cancelled, return approved/rejected
Pattern	Server-side only. Called from Route Handlers — never client-side.
Somali SMS	Twilio supports Somaliland numbers (+252) via international SMS.
Failure	SMS failure is non-blocking. Logged to sms_logs table. Auto-retried once.

```
Install: pnpm add twilio@5.3.5
```

16. PDF Generation — jsPDF

Package	Version	Stars	Purpose
jspdf	2.5.2	■■ 29k	PDF generation engine. Used for: POS receipts, session summaries, return receipts
jspdf-autotable	3.8.4	■■ 4.1k	Table plugin for jsPDF. Generates product tables in receipts and reports.

```
Install: pnpm add jspdf@2.5.2 jspdf-autotable@3.8.4
```

17. File Processing

Package	Version	Stars	Purpose
sharp	0.33.5	■■ 29k	Server-side image processing. Resizes and converts product images to WebP on upload
papaparse	5.4.1	■■ 12k	CSV parsing for bulk product import. Parses the admin-uploaded CSV file into an array

Install:

```
pnpm add sharp@0.33.5 papaparse@5.4.1

pnpm add -D @types/papaparse@5.3.14
```

sharp & Vercel: In next.config.js, add sharp to serverComponentsExternalPackages so it is not bundled by webpack (it uses native binaries): experimental: { serverComponentsExternalPackages: ['sharp'] }

18. Offline Storage — idb

Property	Value
Package	idb
Version	8.0.0
Stars	■■ 6k — No well-established alternative. Created by Jake Archibald (Google Chrome team). The definitive type
Used for	Cart persistence (customer), POS offline transaction queue, product catalog cache for offline search
Pattern	Create one DB per portal in a lib/idb.ts file. Three object stores: cart, pos_queue, catalog_cache.

Install: pnpm add idb@8.0.0

19. Barcode Rendering — react-barcode

Property	Value
Package	react-barcode
Version	1.5.3
Stars	■■ 1.5k — No well-established React barcode alternative. The library is stable, unmaintained but fully functional
Used for	Order number barcode on POS receipts, barcode label printing (50×30mm labels for product shelves)
Format	Code128 (default) — supports alphanumeric SKUs
Output	SVG (for screen display) or canvas (for print)

Install: pnpm add react-barcode@1.5.3

20. Testing

Package	Version	Stars	Purpose
vitest	2.1.3	■ 13k	Unit and integration test runner. Native TypeScript, ESM, fast
@testing-library/react	16.0.1	■ 19k	React component testing. Renders components and queries b
@testing-library/jest-dom	6.6.3	■ 11k	Custom matchers (toBeInTheDocument, toHaveValue, etc.) fo
@testing-library/user-event	14.5.2	■ 7.6k	Simulates real user interactions (type, click, tab) more accurat
jsdom	25.0.1	■ 20k	DOM simulation for Vitest (set testEnvironment: 'jsdom' in vite
@playwright/test	1.48.0	■ 67k	End-to-end browser testing. Tests customer checkout, POS flo

Install:

```
pnpm add -D vitest@2.1.3 @vitejs/plugin-react@4.3.2 jsdom@25.0.1

pnpm add -D @testing-library/react@16.0.1 @testing-library/jest-dom@6.6.3
@testing-library/user-event@14.5.2

pnpm add -D @playwright/test@1.48.0

pnpm exec playwright install # downloads browser binaries
```

21. Developer Experience — ESLint & Prettier

Package	Version	Stars	Purpose
eslint	8.57.1	■ 25k	Linter. Use ESLint 8 (not 9) — ESLint 9 changed the flat config
eslint-config-next	14.2.15	■	Next.js official ESLint config. Includes rules for App Router, ac
@typescript-eslint/parser	8.13.0	■ 15k	Parses TypeScript for ESLint rules.
@typescript-eslint/eslint-plugin	8.13.0	■ 15k	TypeScript-specific lint rules (no-explicit-any, no-unused-vars,
prettier	3.3.3	■ 49k	Code formatter. Enforces consistent style across all TypeScript
prettier-plugin-tailwindcss	0.6.8	■ 11k	Sorts Tailwind class names in a consistent order. Prevents me

Install:

```
pnpm add -D eslint@8.57.1 eslint-config-next@14.2.15

pnpm add -D @typescript-eslint/parser@8.13.0 @typescript-eslint/eslint-plugin@8.13.0

pnpm add -D prettier@3.3.3 prettier-plugin-tailwindcss@0.6.8
```

22. Master Dependency Table

Complete pinned dependency list. Copy directly into package.json. ■■■ = star count exception (see individual section for justification). All packages support Node.js 20 LTS and TypeScript 5.

22.1 Production Dependencies

Package	Exact Version	Category	Stars
next	14.2.15	Framework	■■ 128k
react	18.3.1	Framework	■■ 229k
react-dom	18.3.1	Framework	■■ 229k
@supabase/supabase-js	2.45.4	Database/Auth	■■ 13k
@supabase/ssr	0.5.1	Database/Auth	■
@google/generative-ai	0.21.0	AI	■■ Official
tailwindcss	3.4.14	UI	■■ 83k
autoprefixer	10.4.20	UI/Build	■■ 21k
postcss	8.4.47	UI/Build	■■ 28k
@tailwindcss/forms	0.5.9	UI	■■ 83k
@tailwindcss/typography	0.5.15	UI	■■ 83k
class-variance-authority	0.7.0	UI/shadcn	■■■ 5k
clsx	2.1.1	UI/shadcn	■■■ 8k
tailwind-merge	2.5.4	UI/shadcn	■■ 10k
lucide-react	0.454.0	UI/Icons	■■ 11k
@radix-ui/react-dialog	1.1.2	UI/shadcn	■
@radix-ui/react-select	2.1.2	UI/shadcn	■
@radix-ui/react-tabs	1.1.1	UI/shadcn	■
@radix-ui/react-dropdown-menu	2.1.2	UI/shadcn	■
@radix-ui/react-checkbox	1.1.2	UI/shadcn	■
@radix-ui/react-switch	1.1.1	UI/shadcn	■
@radix-ui/react-slider	1.2.1	UI/shadcn	■
@radix-ui/react-toast	1.2.2	UI/shadcn	■
@radix-ui/react-avatar	1.1.1	UI/shadcn	■
@radix-ui/react-progress	1.1.0	UI/shadcn	■
@radix-ui/react-separator	1.1.0	UI/shadcn	■

Package	Exact Version	Category	Stars
@radix-ui/react-label	2.1.0	UI/shadcn	■
@radix-ui/react-slot	1.1.0	UI/shadcn	■
@radix-ui/react-tooltip	1.1.3	UI/shadcn	■
@radix-ui/react-accordion	1.2.1	UI/shadcn	■
@radix-ui/react-scroll-area	1.2.0	UI/shadcn	■
recharts	2.13.3	Charts	■ 24k
zustand	4.5.5	State	■ 47k
@tanstack/react-query	5.59.0	State	■ 43k
react-hook-form	7.53.1	Forms	■ 41k
zod	3.23.8	Validation	■ 34k
@hookform/resolvers	3.9.1	Forms	■■ 5k
next-intl	3.22.1	i18n	■■ 8k
next-pwa	5.6.0	PWA	■■ 7.8k
@upstash/ratelimit	2.0.4	Rate Limit	■■ 4k
@upstash/redis	1.34.3	Rate Limit	■■ 4k
twilio	5.3.5	Notifications	■■ 3k Official
jspdf	2.5.2	PDF	■ 29k
jspdf-autotable	3.8.4	PDF	■ 4.1k
sharp	0.33.5	Files	■ 29k
papaparse	5.4.1	Files	■ 12k
idb	8.0.0	Storage	■■ 6k
react-barcode	1.5.3	Barcode	■■ 1.5k

22.2 Development Dependencies

Package	Exact Version	Category
typescript	5.6.3	Language
@types/node	20.17.6	Types
@types/react	18.3.12	Types
@types/react-dom	18.3.1	Types
@types/papaparse	5.3.14	Types
eslint	8.57.1	Linting

Package	Exact Version	Category
eslint-config-next	14.2.15	Linting
@typescript-eslint/parser	8.13.0	Linting
@typescript-eslint/eslint-plugin	8.13.0	Linting
prettier	3.3.3	Formatting
prettier-plugin-tailwindcss	0.6.8	Formatting
vitest	2.1.3	Testing
@vitejs/plugin-react	4.3.2	Testing
jsdom	25.0.1	Testing
@testing-library/react	16.0.1	Testing
@testing-library/jest-dom	6.6.3	Testing
@testing-library/user-event	14.5.2	Testing
@playwright/test	1.48.0	Testing
@tanstack/react-query-devtools	5.59.0	DevTools

23. package.json Template

Copy this file to the root of the project. Replace [your-project-name].

```
{
  "name": "[your-project-name]",
  "version": "1.0.0",
  "private": true,
  "engines": {
    "node": ">=20.18.1 <21.0.0",
    "pnpm": ">=9.12.1"
  },
  "scripts": {
    "dev": "next dev",
    "build": "next build",
    "start": "next start",
    "lint": "next lint",
    "format": "prettier --write .",
    "test": "vitest",
    "test:ui": "vitest --ui",
    "test:e2e": "playwright test",
    "type-check": "tsc --noEmit"
  },
  "dependencies": {
    "next": "14.2.15",
    "react": "18.3.1",
    "react-dom": "18.3.1",
    "@supabase/supabase-js": "2.45.4",
    "@supabase/ssr": "0.5.1",
    "@google/generative-ai": "0.21.0",
    "tailwindcss": "3.4.14",
    "autoprefixer": "10.4.20",
    "postcss": "8.4.47",
    "@tailwindcss/forms": "0.5.9",
    "@tailwindcss/typography": "0.5.15",
    "class-variance-authority": "0.7.0",
    "clsx": "2.1.1",
    "tailwind-merge": "2.5.4",
    "lucide-react": "0.454.0",
    "@radix-ui/react-dialog": "1.1.2",
    "@radix-ui/react-select": "2.1.2",
    "@radix-ui/react-tabs": "1.1.1",
    "@radix-ui/react-dropdown-menu": "2.1.2",
    "@radix-ui/react-checkbox": "1.1.2",
```

```

"@radix-ui/react-switch": "1.1.1",
"@radix-ui/react-slider": "1.2.1",
"@radix-ui/react-toast": "1.2.2",
"@radix-ui/react-avatar": "1.1.1",
"@radix-ui/react-progress": "1.1.0",
"@radix-ui/react-separator": "1.1.0",
"@radix-ui/react-label": "2.1.0",
"@radix-ui/react-slot": "1.1.0",
"@radix-ui/react-tooltip": "1.1.3",
"@radix-ui/react-accordion": "1.2.1",
"@radix-ui/react-scroll-area": "1.2.0",
"recharts": "2.13.3",
"zustand": "4.5.5",
"@tanstack/react-query": "5.59.0",
"react-hook-form": "7.53.1",
"zod": "3.23.8",
"@hookform/resolvers": "3.9.1",
"next-intl": "3.22.1",
"next-pwa": "5.6.0",
"@upstash/ratelimit": "2.0.4",
"@upstash/redis": "1.34.3",
"twilio": "5.3.5",
"jspdf": "2.5.2",
"jspdf-autotable": "3.8.4",
"sharp": "0.33.5",
"papaparse": "5.4.1",
"idb": "8.0.0",
"react-barcode": "1.5.3"
},
"devDependencies": {
  "typescript": "5.6.3",
  "@types/node": "20.17.6",
  "@types/react": "18.3.12",
  "@types/react-dom": "18.3.1",
  "@types/papaparse": "5.3.14",
  "eslint": "8.57.1",
  "eslint-config-next": "14.2.15",
  "@typescript-eslint/parser": "8.13.0",
  "@typescript-eslint/eslint-plugin": "8.13.0",
  "prettier": "3.3.3",
  "prettier-plugin-tailwindcss": "0.6.8",
  "vitest": "2.1.3",
  "@vitejs/plugin-react": "4.3.2",
  "jsdom": "25.0.1",
  "@testing-library/react": "16.0.1",

```

```
"@testing-library/jest-dom": "6.6.3",  
"@testing-library/user-event": "14.5.2",  
"@playwright/test": "1.48.0",  
"@tanstack/react-query-devtools": "5.59.0"  
}  
}
```


24. Configuration Files

24.1 tsconfig.json

```
{
  "compilerOptions": {
    "target": "ES2017",
    "lib": ["dom", "dom.iterable", "esnext"],
    "allowJs": true,
    "skipLibCheck": true,
    "strict": true,
    "noEmit": true,
    "esModuleInterop": true,
    "module": "esnext",
    "moduleResolution": "bundler",
    "resolveJsonModule": true,
    "isolatedModules": true,
    "jsx": "preserve",
    "incremental": true,
    "plugins": [{ "name": "next" }],
    "paths": { "@/*": [".src/*"] }
  },
  "include": ["next-env.d.ts", "**/*.ts", "**/*.tsx", ".next/types/**/*.ts"],
  "exclude": ["node_modules"]
}
```

24.2 tailwind.config.ts

```
import type { Config } from 'tailwindcss'
import { fontFamily } from 'tailwindcss/defaultTheme'

const config: Config = {
  darkMode: ['class'],
  content: [
    './src/pages/**/*.{js,ts,jsx,tsx,mdx}',
    './src/components/**/*.{js,ts,jsx,tsx,mdx}',
    './src/app/**/*.{js,ts,jsx,tsx,mdx}',
  ],
  theme: {
    extend: {
      colors: {
        brand: {
          50: '#FFF7ED', 100: '#FFEDD5', 200: '#FED7AA',
          300: '#FDBA74', 400: '#FB923C', 500: '#F97316',

```

```

600: '#EA580C', 700: '#C2410C', 800: '#9A3412',
900: '#7C2D12',
},
},
fontFamily: { sans: ['var(--font-inter)', ...fontFamily.sans] },
},
},
plugins: [
  require('@tailwindcss/forms'),
  require('@tailwindcss/typography'),
],
}
export default config

```

24.3 next.config.js (use .js not .ts — required by next-pwa)

```

/** @type {import('next').NextConfig} */
const withPWA = require('next-pwa')({
  dest: 'public',
  disable: process.env.NODE_ENV === 'development',
  register: true,
  skipWaiting: true,
  runtimeCaching: require('next-pwa/cache'),
})
const nextConfig = {
  images: {
    remotePatterns: [{
      protocol: 'https',
      hostname: '*.supabase.co',
      pathname: '/storage/v1/object/public/**',
    }],
  },
  experimental: {
    serverComponentsExternalPackages: ['sharp', 'twilio'],
  },
}
module.exports = withPWA(nextConfig)

```

24.4 postcss.config.js

```

module.exports = {
  plugins: {
    tailwindcss: {},

```

```
autoprefixer: {},  
},  
}
```

24.5 .eslintrc.json

```
{  
  "extends": [  
    "next/core-web-vitals",  
    "plugin:@typescript-eslint/recommended"  
  ],  
  "parser": "@typescript-eslint/parser",  
  "plugins": ["@typescript-eslint"],  
  "rules": {  
    "@typescript-eslint/no-unused-vars": ["error", { "argsIgnorePattern": "^_" }],  
    "@typescript-eslint/no-explicit-any": "error",  
    "@typescript-eslint/consistent-type-imports": "error",  
    "prefer-const": "error"  
  }  
}
```

24.6 .prettierrc

```
{  
  "semi": false,  
  "singleQuote": true,  
  "tabWidth": 2,  
  "trailingComma": "es5",  
  "printWidth": 100,  
  "plugins": ["prettier-plugin-tailwindcss"]  
}
```

24.7 .nvmrc (pins Node.js version for nvm users)

```
20.18.1
```

24.8 vitest.config.ts

```
import { defineConfig } from 'vitest/config'  
import react from '@vitejs/plugin-react'  
import { resolve } from 'path'
```

```
export default defineConfig({
  plugins: [react()],
  test: {
    environment: 'jsdom',
    globals: true,
    setupFiles: ['./src/test/setup.ts'],
    coverage: {
      provider: 'v8',
      reporter: ['text', 'json', 'html'],
      exclude: ['node_modules', '.next', 'src/test'],
    },
  },
  resolve: {
    alias: { '@': resolve(__dirname, './src') },
  },
})
```

src/test/setup.ts — imported by vitest.config.ts before each test:

```
import '@testing-library/jest-dom'
```

25. Environment Variables Reference

Create a `.env.local` file at the project root (never commit to git). Add each variable to the Vercel dashboard under Settings → Environment Variables.

.gitignore must include: `.env .env.local .env.*.local`

Use `.env.example` (with blank values) as a documentation file that is committed to git.

Variable	Used In	Description
NEXT_PUBLIC_SUPABASE_URL	Client + Server	Supabase project URL. Format: <code>https://[ref].supabase.co</code>
NEXT_PUBLIC_SUPABASE_ANON_KEY	Client + Server	Supabase anonymous (public) key. Safe to expose to browser — RLS protection
SUPABASE_SERVICE_ROLE_KEY	Server only	Supabase service role key. Bypasses RLS. Never expose to client. Used for
GEMINI_API_KEY	Server only	Google AI Studio API key. Get from: <code>aistudio.google.com</code>
UPSTASH_REDIS_REST_URL	Server only	Upstash Redis database REST URL. From Upstash console → Database → REST A
UPSTASH_REDIS_REST_TOKEN	Server only	Upstash Redis REST token. From Upstash console → Database → REST A
TWILIO_ACCOUNT_SID	Server only	Twilio Account SID. From Twilio console → Account Info.
TWILIO_AUTH_TOKEN	Server only	Twilio Auth Token. From Twilio console → Account Info.
TWILIO_PHONE_NUMBER	Server only	Twilio sender phone number. Format: <code>+1XXXXXXXXXX</code>
NEXT_PUBLIC_APP_URL	Client + Server	Full URL of the deployed app. Example: <code>https://borama-hardware.vercel.app</code>

25.1 .env.example Template

```
# Supabase
NEXT_PUBLIC_SUPABASE_URL=
NEXT_PUBLIC_SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY=

# Google Gemini AI
GEMINI_API_KEY=

# Upstash Redis (rate limiting)
UPSTASH_REDIS_REST_URL=
UPSTASH_REDIS_REST_TOKEN=

# Twilio SMS
TWILIO_ACCOUNT_SID=
TWILIO_AUTH_TOKEN=
TWILIO_PHONE_NUMBER=

# App
NEXT_PUBLIC_APP_URL=http://localhost:3000
```

25.2 Validate Environment Variables at Startup

Add this to `src/lib/env.ts` and import it in `src/app/layout.tsx` to crash the build early if any required variable is missing:

```
import { z } from 'zod'
const envSchema = z.object({
  NEXT_PUBLIC_SUPABASE_URL: z.string().url(),
  NEXT_PUBLIC_SUPABASE_ANON_KEY: z.string().min(1),
  SUPABASE_SERVICE_ROLE_KEY: z.string().min(1),
  GEMINI_API_KEY: z.string().min(1),
  UPSTASH_REDIS_REST_URL: z.string().url(),
  UPSTASH_REDIS_REST_TOKEN: z.string().min(1),
  TWILIO_ACCOUNT_SID: z.string().startsWith('AC'),
  TWILIO_AUTH_TOKEN: z.string().min(1),
  TWILIO_PHONE_NUMBER: z.string().startsWith('+'),
  NEXT_PUBLIC_APP_URL: z.string().url(),
})
export const env = envSchema.parse(process.env)
```

Borama Hardware Store
Tech Stack & Dependency Specification
Borama, Somaliland

All 53 packages pinned to exact versions.

Validated against Node.js 20.18.1 LTS.

Version 1.0 — July 2025